Q12:- Let $T_A: \mathbb{R}^2 \to \mathbb{R}^2$ be multiplication by A. Determine whether the vector $\underline{u} = (1,2)$ is in the span of $\{T_A(\underline{e}_1), T_A(\underline{e}_2)\}$.

a. $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

$\underline{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\underline{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

So $T_A(\underline{e}_1) = A\underline{e}_1 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ (1st column of A)

& $T_A(\underline{e}_2) = A\underline{e}_2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ (2nd column of A)

To check that $\underline{u} = (1,2)$ is in span of $\{T_A(\underline{e}_1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, T_A(\underline{e}_2) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}\}$, it suffices

$$k_1\begin{bmatrix} 1 \\ 0 \end{bmatrix} + k_2\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{—(1)}$$

for some scalars $k_1, k_2 \in \mathbb{R}$.

The augmented matrix of (1) is

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \end{bmatrix} \begin{matrix} R_1 - 2R_2 \end{matrix}$$

$\therefore k_1 = -3$ and $k_2 = 2$

Because there exists scalars such that (1) is satisfied, so $\underline{u}$ lies in the span of $\{T_A(\underline{e}_1), T_A(\underline{e}_2)\}$.

Q14:- Let $f = \cos^2 x$ and $g = \sin^2 x$. Which of the following lie in the space spanned by f and g?

a. $\cos 2x$ b. $3 + x^2$ c. 1 d. $\sin x$

e. 0

a. Since $\cos 2x = \cos^2 x - \sin^2 x = f - g$

Thus $\cos 2x$ can be written as a linear combination of f and g. Thus $\cos 2x$ lies the span of f and g.

b. Since $k_1 \cos^2 x + k_2 \sin^2 x \neq 3 + x^2$

for any set of scalars k_1, k_2, so $3 + x^2$ does not belong to the space spanned by f and g.

c. Since $\cos^2 x + \sin^2 x = 1$

or $f + g = 1$

so 1 lies in the span of f and g.

d. Since $k \cos^2 x + k \sin^2 x \neq \sin x$

for any set of scalars k_1, k_2, so $\sin x$ is not in the span of f and g.

e. $\because 0 \cos^2 x + 0 \sin^2 x = 0$ or $0f + 0g = 0$

so 0 lies in the span of f and g.

Q.5:- Let W be the solution space to the system $Ax = 0$. Determine whether the set $\{\underline{u}, \underline{v}\}$ spans W.

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

b. $\underline{u} = (1, 0, -1, 0)$

$\underline{v} = (1, 1, -1, -1)$

First find the solution space of $A\underline{x} = 0$

The coefficient matrix is

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -1 & 0 & -1 \\ 0 & 1 & 0 & 1 \end{bmatrix} R_2 - R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_3 + R_2 \sim \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_1 + R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} -R_2$$

The system $A\underline{x} = \underline{0}$ is equivalent to

$x_1 + x_3 = 0$

$x_2 + x_4 = 0$

Free variables $= x_3, x_4$

Let $x_3 = s$ $\quad x_4 = t$

$x_1 = -s$ $\quad x_2 = -t$

∴ The solution set of $A\underline{x} = \underline{0}$ is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -s \\ -t \\ s \\ t \end{bmatrix}$$

$$= \begin{bmatrix} -s+0t \\ 0s-t \\ s+0t \\ 0s+t \end{bmatrix} = \begin{bmatrix} -s \\ 0s \\ s \\ 0s \end{bmatrix} + \begin{bmatrix} 0t \\ -t \\ 0t \\ t \end{bmatrix}$$

$$= s\begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

∴ solution set of $A\underline{x} = \underline{0}$ is spanned by $\left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}$

So $W = \text{span}\left\{\begin{bmatrix}-1\\0\\0\\1\end{bmatrix}, \begin{bmatrix}0\\-1\\0\\1\end{bmatrix}\right\}$

Now to show that $\{\underline{u}, \underline{v}\}$ span W, it is necessary to show that $\begin{bmatrix}-1\\0\\0\\1\end{bmatrix}$ and $\begin{bmatrix}0\\-1\\0\\1\end{bmatrix}$ lie in the span of $\underline{u}$ and $\underline{v}$.

To this end:

construct the augmented matrix to solve the system $k_1\underline{u} + k_2\underline{v} = w_i$ $i=1,2$ simultaneously, where

$$w_1 = \begin{bmatrix}-1\\0\\0\\1\end{bmatrix} \text{ and } w_2 = \begin{bmatrix}0\\-1\\0\\1\end{bmatrix}.$$

The augmented matrix is

$$\left[\begin{array}{cc|cc} 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{array}\right] \quad R_3 + R_1$$

$$\left[\begin{array}{cc|cc} 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad R_4 + R_2$$

Since the system is consistent for both vectors, so w_1, w_2 belong to the span of $\underline{u}$ and $\underline{v}$ and hence

The set $\{u, v\}$ spans W.

Q18:- Let $T_A: \mathbb{R}^3 \to \mathbb{R}^2$ be multiplication by A and let $u_1 = (0, 1, 1)$ and $u_2 = (2, -1, 1)$ and $u_3 = (1, 1, -2)$. Determine whether the set $\{T_A(u_1), T_A(u_2), T_A(u_3)\}$ spans $\mathbb{R}^2$.

b. $A = \begin{bmatrix} 0 & 0 \\ 1 & -3 \\ 1 & \end{bmatrix}$

$$T_A(u_1) = A u_1 = \begin{bmatrix} 0 & 0 \\ 1 & -3 \\ 1 & \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ -2 \end{bmatrix}$$

$$T_A(u_2) = A u_2 = \begin{bmatrix} 0 & 0 \\ 1 & -3 \\ 1 & \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ 2-1-3 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$$

$$T_A(u_3) = A u_3 = \begin{bmatrix} 0 & 0 \\ 1 & -3 \\ 1 & \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1+1+6 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$$

To show that $\{T_A(u_1) = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$, $T_A(u_2) = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$, $T_A(u_3) = \begin{bmatrix} 1 \\ 8 \end{bmatrix}\}$ spans $\mathbb{R}^2$, it suffices every vector $(a, b) \in \mathbb{R}^2$ can be written as linear combination of $\begin{bmatrix} 0 \\ -2 \end{bmatrix}$, $\begin{bmatrix} -1 \\ -2 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 8 \end{bmatrix}$

i.e. $k_1\begin{bmatrix}0\\-2\end{bmatrix}+k_2\begin{bmatrix}-1\\-2\end{bmatrix}+k_3\begin{bmatrix}1\\8\end{bmatrix}=\begin{bmatrix}a\\b\end{bmatrix}$

The augmented matrix is

$$\begin{bmatrix}0 & -1 & 1 & ; & a\\ -2 & -2 & 8 & ; & b\end{bmatrix}$$

$$\begin{bmatrix}0 & -1 & 1 & ; & a\\ 1 & 1 & -4 & ; & -\frac{b}{2}\end{bmatrix}\ -\frac{1}{2}R_2$$

$$\begin{bmatrix}1 & 1 & -4 & ; & -\frac{b}{2}\\ 0 & -1 & 1 & ; & a\end{bmatrix}\ R_{12}$$

$$\begin{bmatrix}1 & 1 & -4 & ; & -\frac{b}{2}\\ 0 & -1 & 1 & ; & -a\end{bmatrix}\ -R_2$$

which is consistent for all choices of a and b, so

$\{T_A(\underline{u}_1), T_A(\underline{u}_2), T_A(\underline{u}_3)\}$ Spans $\mathbb{R}^2$.

Q20:- Let $\underline{v}_1=(1,6,4)$, $\underline{v}_2=(2,4,-1)$, $\underline{v}_3=(-1,2,5)$, and $\underline{w}_1=(1,-2,-5)$ $\underline{w}_2=(0,8,9)$. Show that

$$\text{Span}\{\underline{v}_1,\underline{v}_2,\underline{v}_3\}=\text{Span}\{\underline{w}_1,\underline{w}_2\}$$

First it will be shown that

$$\text{Span}\{\underline{v}_1,\underline{v}_2,\underline{v}_3\}\subseteq\text{Span}\{\underline{w}_1,\underline{w}_2\}\ \textcircled{1}$$

To show ①, it will be shown that $\underline{v}_1,\underline{v}_2,\underline{v}_3$ all belongs to span of $\underline{w}_1,\underline{w}_2$.

i.e. There exist scalars such that

$$k_1 \underline{w}_1 + k_2 \underline{w}_2 = \underline{v}_i \qquad i = 1, 2, 3$$

The augmented matrix is

$$\left[\begin{array}{cc|ccc} 1 & 0 & 1 & 2 & -1 \\ -2 & 8 & 6 & 4 & 2 \\ -5 & 9 & 4 & -1 & 5 \end{array}\right]$$

$$\left[\begin{array}{cc|ccc} 1 & 0 & 1 & 2 & -1 \\ 0 & 8 & 8 & 8 & 0 \\ 0 & 9 & 9 & 9 & 0 \end{array}\right] \begin{array}{l} R_2 + 2R_1 \\ R_3 + 5R_1 \end{array}$$

$$\left[\begin{array}{cc|ccc} 1 & 0 & 1 & 2 & -1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{array}\right] \begin{array}{l} \frac{1}{8} R_2 \\ \frac{1}{9} R_3 \end{array}$$

$$\left[\begin{array}{cc|ccc} 1 & 0 & 1 & 2 & -1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] R_3 - R_2$$

Because it is consistent for all $\underline{v}_1, \underline{v}_2, \underline{v}_3$, so $\underline{v}_1, \underline{v}_2, \underline{v}_3$ lie in span of $\underline{w}_1, \underline{w}_2$.

$$\therefore \quad \text{span}\{\underline{v}_1, \underline{v}_2, \underline{v}_3\} \subseteq \text{span}\{\underline{w}_1, \underline{w}_2\}$$

Now the converse will be shown

i.e.

$$\text{span}\{\underline{w}_1, \underline{w}_2\} \subseteq \text{span}\{\underline{v}_1, \underline{v}_2, \underline{v}_3\} \quad -(2)$$

Need to show that there exist scalar k_1, k_2, k_3 such that

$$k_1 \underline{v}_1 + k_2 \underline{v}_2 + k_3 \underline{v}_3 = \underline{w}_i \qquad i = 1, 2$$

The augmented matrix is

$$\left(\begin{array}{ccc|cc} 1 & 2 & -1 & 1 & 0 \\ 6 & 4 & 2 & -2 & 8 \\ 4 & -1 & 5 & -5 & 9 \end{array}\right)$$

$$\left(\begin{array}{ccc|cc} 1 & 2 & -1 & 1 & 0 \\ 0 & -8 & 8 & -8 & 8 \\ 0 & -9 & 9 & -9 & 9 \end{array}\right) \begin{array}{l} R_2 - 6R_1 \\ R_3 - 4R_1 \end{array}$$

$$\left(\begin{array}{ccc|cc} 1 & 2 & -1 & 1 & 0 \\ 0 & 1 & -1 & 1 & -1 \\ 0 & 1 & -1 & 1 & -1 \end{array}\right) \begin{array}{l} -\frac{1}{8} R_2 \\ -\frac{1}{9} R_3 \end{array}$$

$$\left(\begin{array}{ccc|cc} 1 & 2 & -1 & 1 & 0 \\ 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) R_3 + R_2$$

This is consistent as well, so $\underline{w}_1, \underline{w}_2$ lie in span of $\underline{v}_1, \underline{v}_2$ and $\underline{v}_3$, thus

$$\text{span}\{\underline{w}_1, \underline{w}_2\} \subseteq \text{span}\{\underline{v}_1, \underline{v}_2, \underline{v}_3\}$$

Thus from ① and ②

$$\text{span}\{\underline{v}_1, \underline{v}_2, \underline{v}_3\} = \text{span}\{\underline{w}_1, \underline{w}_2\}.$$

Q22: Let V be the solution space of the equation $4x - y + 2z = 0$ and let W be the subspace of $\mathbb{R}^3$ spanned by $(1,1,1)$. Find a vector $\underline{v}$ in V and a vector $\underline{w}$ in W for which

$$\underline{v} + \underline{w} = (1,0,1)$$

First, vectors spanning V will be found

$$4x - y + 2z = 0$$

Free variables $= y, z$

let $y = s$, $z = t$

So $4x = y - 2z$

$\Rightarrow 4x = s - 2t$

$\Rightarrow x = \frac{1}{4}s - \frac{1}{2}t$

Thus solution space in vector form:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{1}{4}s - \frac{1}{2}t \\ s \\ t \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{4}s - \frac{1}{2}t \\ s + 0t \\ 0s + t \end{bmatrix}$$

$$= s\begin{bmatrix} 1/4 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} -1/2 \\ 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{4}s\begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix} + \frac{1}{2}t\begin{bmatrix} -1 \\ 0 \\ +2 \end{bmatrix}$$

$$\therefore V = \text{Span}\left\{\underline{v}_1 = \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix}, \underline{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}\right\}$$

Let $k_1, k_2, k_3 \in \mathbb{R}$ such that

$$k_1 v_1 + k_2 v_2 + k_3 w_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

where $w_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 4 & 0 & 1 & 0 \\ 0 & 2 & 1 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 4 & -3 & -4 \\ 0 & 2 & 1 & 1 \end{array}\right] \quad R_2 - 4R_1$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 0 & -5 & -6 \\ 0 & 2 & 1 & 1 \end{array}\right] \quad R_2 - 2R_3$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & -5 & -6 \end{array}\right] \quad R_{23}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{6}{5} \end{array}\right] \quad \frac{1}{2}R_2 \quad -\frac{1}{5}R_3$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 0 & -1/5 \\ 0 & 1 & 0 & -1/10 \\ 0 & 0 & 1 & 6/5 \end{array}\right) \quad R_2 - \frac{1}{2}R_3$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & -3/10 \\ 0 & 1 & 0 & -1/10 \\ 0 & 0 & 1 & 6/5 \end{array}\right) \quad R_1 + R_2$$

$\frac{1}{2} - \frac{3}{5}$

$\frac{5-6}{10}$

$-\frac{1}{5} - \frac{1}{10}$

$\frac{-2-1}{10}$

$$\therefore \frac{-3}{10}\begin{pmatrix}1\\4\\0\end{pmatrix} - \frac{1}{10}\begin{pmatrix}-1\\0\\2\end{pmatrix} + \frac{6}{5}\begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}1\\0\\1\end{pmatrix}$$

Thus $\underline{v} = \frac{-3}{10}\begin{pmatrix}1\\4\\0\end{pmatrix} - \frac{1}{10}\begin{pmatrix}-1\\0\\2\end{pmatrix}$

$$= \begin{pmatrix}\frac{-3}{10} + \frac{1}{10}\\ -\frac{12}{10} + 0 \\ 0 - \frac{2}{10}\end{pmatrix} = \begin{pmatrix}-1/5\\ -6/5 \\ -1/5\end{pmatrix} \in V$$

$$\underline{w} = \frac{6}{5}\begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}6/5\\6/5\\6/5\end{pmatrix} \in W$$